

TEBEA, Romania

WMO: 152060

Lat: 46.1694N

Lon: 22.7261E

Elev: 900

StdP: 14.22

Time Zone: 2.00 (EUE)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 10.1	(c) 14.5	(d) 5.1	(e) 7.4	(f) 13.2	(g) 10.0	(h) 9.5	(i) 17.5	(j) 16.3	(k) 35.1	(l) 13.0	(m) 34.0	(n) 2.5	(o) 90	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 23.6	(c) 89.5	(d) 69.1	(e) 86.2	(f) 68.9	(g) 83.0	(h) 67.9	(i) 71.8	(j) 84.1	(k) 70.4	(l) 82.5	(m) 69.2	(n) 80.7	(o) 4.9	(p) 120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.0	106.4	76.2	66.7	101.5	74.8	65.5	97.2	73.4	36.0	84.8	34.8	82.5	33.8	80.8	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 13.9	(b) 11.6	(c) 9.7	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 13.9	(b) 11.6	(c) 9.7	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.8	29.5	35.6	41.9	51.2	59.6	66.4	68.0	69.0	60.0	51.2	42.5	33.7		
	DBStd	15.07	8.15	7.34	7.06	8.22	5.44	5.66	5.01	5.51	7.31	6.88	7.65	6.67		
	HDD50	2221	637	405	264	86	3	0	0	0	5	71	245	505		
	HDD65	5595	1102	824	716	414	181	50	28	24	183	427	676	970		
	CDD50	2509	0	1	13	121	300	492	558	589	306	110	19	0		
	CDD65	408	0	0	0	0	13	92	122	148	33	0	0	0		
	CDH74	4582	0	0	2	70	223	943	1307	1550	439	48	0	0		
	CDH80	1490	0	0	0	2	26	270	442	624	124	2	0	0		
	WSAvg	3.4	2.8	3.4	3.8	4.4	3.6	3.6	3.7	3.4	3.1	3.2	3.2	3.1		
	PrecAvg	27.2	1.7	1.4	1.6	2.2	2.8	3.3	3.3	2.0	2.6	2.3	2.0	2.1		
	PrecMax	36.0	3.0	4.6	3.8	4.3	5.4	6.5	6.6	4.3	5.9	5.2	4.2	5.3		
	PrecMin	18.7	0.6	0.2	0.4	0.4	1.3	0.9	0.8	0.5	0.2	0.1	0.6	0.1		
	PrecStd	4.6	0.8	1.0	1.0	0.9	1.0	1.4	1.7	1.0	1.3	1.3	0.9	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.0	59.3	71.5	79.5	83.6	89.2	92.2	94.5	88.8	78.6	67.0	53.5		
		MCWB	45.8	49.0	54.2	60.7	65.5	70.9	69.9	70.3	67.0	59.8	54.9	46.9		
	2%	DB	45.6	54.7	65.1	75.8	79.5	86.1	88.6	90.5	83.3	74.4	62.6	48.8		
		MCWB	42.2	46.3	50.9	58.0	64.2	70.2	69.3	68.8	66.1	59.2	53.2	44.8		
	5%	DB	42.1	50.3	59.5	71.8	76.2	82.9	84.6	86.4	79.1	69.5	58.0	45.0		
		MCWB	40.1	43.8	48.4	55.9	62.3	68.5	69.0	68.6	64.2	57.8	50.8	42.7		
	10%	DB	38.9	46.2	54.4	67.1	72.5	79.9	81.2	82.5	74.2	63.9	54.2	42.0		
		MCWB	37.0	41.7	45.4	54.0	60.7	67.5	67.2	68.0	62.7	55.4	49.3	40.1		
	0.4%	WB	46.9	51.0	55.1	62.6	68.3	74.1	72.8	73.5	69.7	63.0	56.3	48.8		
		MCDB	50.7	58.2	70.0	77.1	79.9	86.4	84.5	87.9	84.8	72.8	63.2	52.7		
	2%	WB	43.1	47.6	52.3	59.4	65.8	71.2	71.3	71.6	67.3	60.2	54.5	45.0		
		MCDB	45.2	53.1	63.7	73.1	76.5	82.6	83.6	84.5	80.4	70.4	60.9	48.1		
	5%	WB	40.4	45.1	49.4	57.4	64.0	69.8	70.1	70.1	65.4	58.5	52.6	42.9		
		MCDB	42.2	49.7	58.6	69.7	73.5	81.4	82.1	82.3	75.9	68.0	57.6	45.8		
	10%	WB	37.4	42.2	46.8	55.3	62.1	68.2	68.8	68.8	63.7	56.4	50.3	40.6		
		MCDB	38.9	45.9	52.9	65.6	70.7	77.8	79.4	80.5	72.2	63.4	54.4	42.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.8	16.2	20.0	22.1	21.1	21.8	23.0	23.6	21.6	21.5	16.2	11.1		
		MCDBR	16.2	23.7	30.4	31.2	28.2	26.9	29.7	31.2	30.2	32.3	23.2	16.3		
		MCWBR	11.7	15.1	16.8	15.0	14.1	12.3	12.5	11.9	13.1	17.6	14.2	12.1		
		MCDBR	15.1	21.3	27.6	27.9	26.0	25.3	25.6	27.0	26.3	28.2	19.3	16.3		
	5% WB	MCWBR	11.0	14.0	15.4	13.6	13.6	12.0	11.6	11.7	12.1	15.5	11.8	12.0		
(40) Clear-Sky Solar Irradiance	taub		0.320	0.349	0.392	0.447	0.434	0.427	0.445	0.452	0.414	0.376	0.346	0.322		
	taud		2.345	2.278	2.216	2.088	2.184	2.244	2.202	2.178	2.252	2.353	2.387	2.380		
	Ebn at Noon		243	257	262	257	265	267	260	252	252	245	231	229		
	Edh at Noon		24	31	39	48	45	43	44	43	37	29	23	21		
(44) All-Sky Solar Radiation	RadAvg		394	647	1006	1374	1648	1833	1798	1654	1177	814	472	316		
	RadStd		53	83	144	145	156	163	157	161	130	96	62	41		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (65 neighbors)		+1.26	N/A	N/A	+1.75	N/A	N/A	-225	-340	+218	+108

Nomenclature: See separate page